GLOBAL CASE STUDY COMPETITION

A finance and consulting competition



Executive Summary

Problem & Insight

- Markets are **intent blind**.
- Financial decisions **often irrational**.
- Driven by Psychological states (Panic selling, Overconfidence).

Result:- Causes major losses.

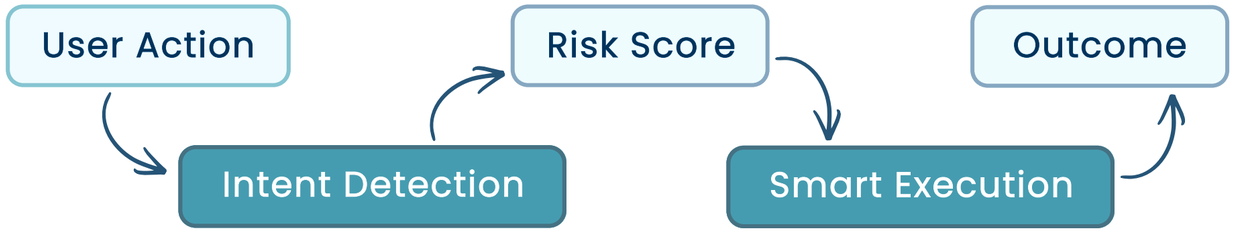


Emotionally Driven Returns



Rational Returns

How it Works





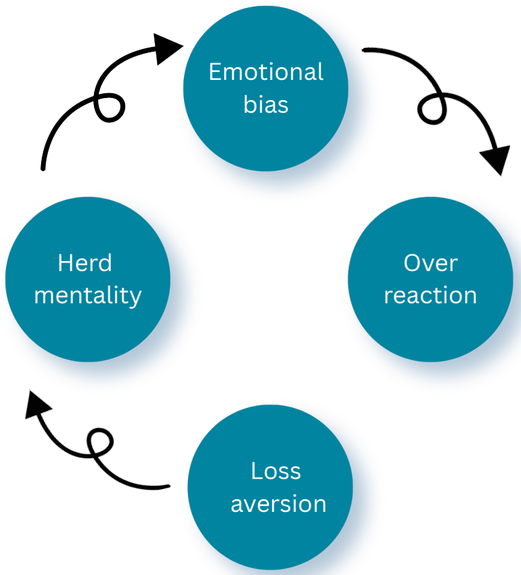
Solution

- **Detects user intent before execution**
- **Identifies emotional triggers in real-time**
- **Intervenes using risk scoring + behavioral insights**

Impact

- **Emotional Trading Losses ↓**
- **Decision Quality & Discipline ↑**
- **Trust & Platform retention ↑**

Emotional Analysis Framework

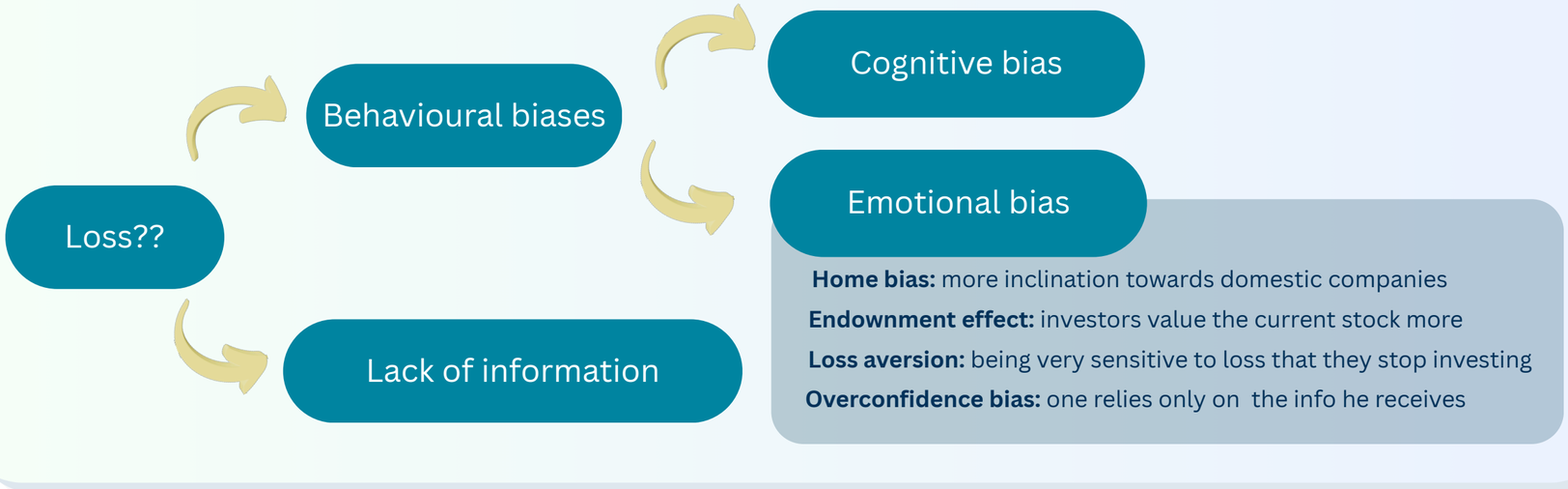


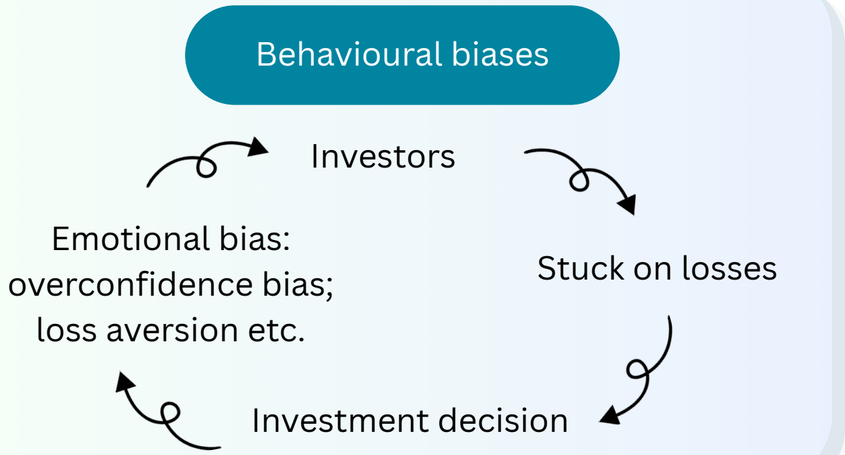
Role of emotions in financial decision making

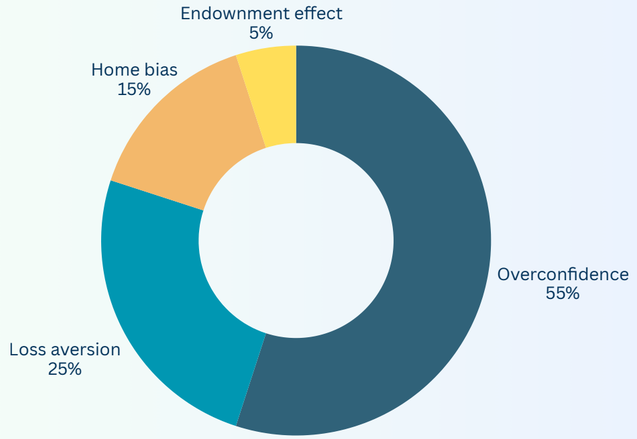
..**Fear & Greed:** greed can lead to excessive risk taking ,fear causes panic selling during

..**Overconfidence bias**: Investors may overestimate their ability and may lead to risky deal.

..**Loss aversion:** people tend to feel loss more accutely than pleasure of gain.







Biases that impact financial decisions

Real World Instances

	2008 CRISIS	COVID 19	CRYPTO
MAIN EMOTIONS INVOLVED	Greed Overconfidence Fear and panic	Panic Fear Uncertainity	Greed Hype & FOMO Overconfidence
PHASE 1	1..Banks gave loan to risky borowers 2..Subprime mortgages increased from *2.5% to nearly 15% of all home loans . 3. Investors kept buying mortage backed securities believing that it was safe .	1..The S&P 500 fell *34% in just 33 days 2. Global stock markets lost *$30 trillion in value in weeks. 3. Volatility index (VIX) reached *85 (highest since 2008 crisis)	1. Bitcoin rose $7,000 → $69,000 2. Crypto market cap reached $3 trillion. 3. Leverage trading increased
PHASE 2	1..Subprime borrowers began defaulting in *early 2007* & mortage backed value collapsed 2.Bankruptcy of Lehman Brothers in September 2008 triggered market crash and bank runs*	1. Investors withdrew *$326 billion from global equity funds* in March 2020 2. Gold prices surged as investors looked for safe assets 3. Trading volumes increased drastically	1. Meme coins surged 2. Leverage trading increased 3. this shows the Overconfidence and the Risk-taking consequences.
PHASE 3	1. Credit markets froze 2. Central banks had to inject liquidity globally 3. Banks stopped trusting each other	1. Consumer confidence dropped sharply worldwide 2. Retail trading accounts surged globally 3. Investment activity slowed in early lockdown period	1. Bitcoin fell $69,000 → $16,000 2. $2 trillion wiped 3. this was followed by the flow of emotion like panic and fear.

Financial Ecosystem Analysis

Core Insight

- Emotional bias doesn't affect all financial sectors equally. It is most damaging where decisions are **frequent**, **high risk** and **time sensitive**.



	Emotional Intensity	Decision Frequency	Impact Level
Retail Banking	Medium	High	Medium
Insurance	Low	Low	Low
Investment & Trading	Very High	High	Very High
Lending & Credit	High	Medium	High
Personal Finance Apps	Medium	High	Medium

Srategic Comparision

Traditional Platforms	Emotion Aware AI
• Automated Transaction • Historical Data • Trust via Performance	• Improved Decision Quality • Real time Intent & Emotion • Trust via Empathy & Safety



Why this Domain?

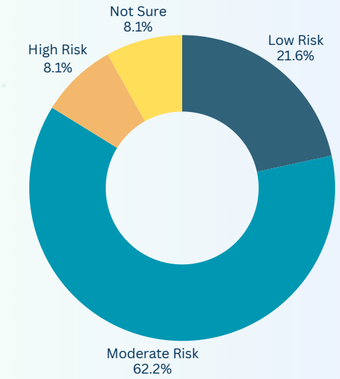
- Maximum emotional distortion.
- Highest financial impact.
- Immediate consequences.
- Strongest opportunity for AI intervention.

User Intent → Emotionally Blind AI → Execution Only (Panic Sell) → Performance Gap

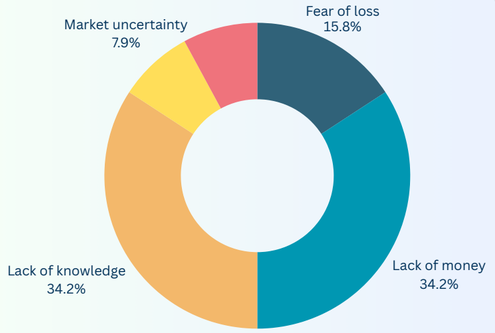
User Research about Emotional Impact on Finance

Retail investors frequently make suboptimal investment decisions due to emotional biases such as fear, greed, and overconfidence. These behaviors lead to panic selling, impulsive buying, and deviation from long-term financial goals, ultimately resulting in lower returns and reduced trust in financial platforms.

USER RESEARCH



User Risk Investment Ratio



Investment Hurdles

KEY FINDING

"78% of investors admit making at least one emotionally-driven financial decision."

"These emotional triggers translate into measurable behavioral risks."

IMPACTS

- Biggest wealth destruction during crashes
- Buying at peaks leads to consistent losses
- Leads to repeated risky decisions
- Capital stuck, opportunity loss
- Investors are not purely rational

Behavioral risk is highly concentrated

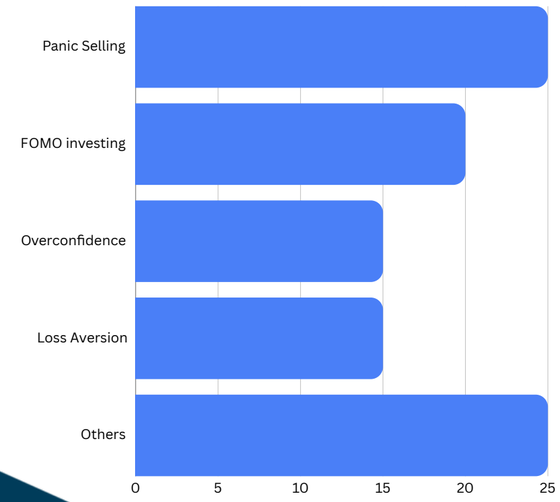
Key Insights

Top 4 behaviors = ~75% of total risk

Panic Selling + FOMO alone = 45% impact

"Value destruction in investing is not evenly distributed — it is driven by a few high-impact emotional behaviors".

Biggest contribution



Panic Selling (25%)

- **Trigger:** Market crash
- **Behavior**: Immediate sell-off
- **Impact:**
  Locks losses
  Misses recovery
- **Stat:**
  ~60% retail investors sell during downturns

Loss Aversion (15%)

- **Trigger:** Fear of realizing loss
- **Behavior:** Holding bad assets
- **Impact:** Capital blockage
- Opportunity loss

FOMO Investing (20%)

- **Trigger:** Market hype / trends
- **Behavior:** Buy at peak
- **Impact:** Overvaluation entry
- High loss probability
- **Stat:**
  70% of trending asset buyers enter at peak price zones

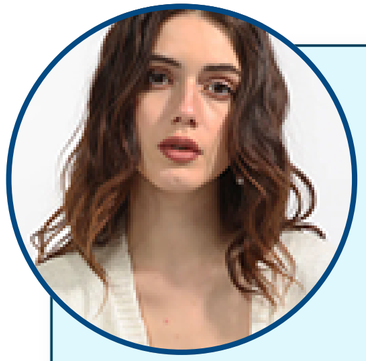
Overconfidence (15%)

- **Trigger:** Past success
- Behavior: High-risk bets
- **Impact:** Poor diversification
- Volatility ↑
- **Stat:**
  Active traders underperform market by ~6–7% annually

"Investor decisions are not just data-driven, but emotion-driven — with over 70% influenced by psychological factors. This creates a significant behavioral risk layer that traditional financial systems fail to address."

Area	Key Insight	Data (%)	Impact
Emotional Decisions	Investors rely on emotions over rational analysis	~70%	Poor timing & inconsistent decisions
Fear & Loss Aversion	Strong fear during volatility leads to panic selling	80%+	Losses locked, missed recovery
Informal Advice	Heavy dependence on friends/social media	60–74%	Herd behavior & misinformation
Financial Planning Gap	Few investors follow structured plans	20–22%	Weak long-term wealth creation
Overconfidence	Investors overestimate market knowledge	~64%	Risky trades & volatility
Social Influence	Decisions shaped by peer pressure	High (qualitative)	Reduced independent thinking

User Persona



TheBeginner
Aanya Sharma
Profile:
Age: 20–30
- New to Stock market
- Limited financial knowledge

Behavior:
- Gets scared during market drops
- Avoids investing or panic sells early

Pain Points:
- Doesn’t understand volatility
- Relies on news/social media
- Lacks confidence

Need:
- Guidance + reassurance + confidence building



The Active Trader (Impulse-Driven)
Rupes Mehta
Profile:
Age: 22–35
Trades frequently
Uses apps daily

Behavior:
- Overtrades
- Chases trends (FOMO)
- Reacts instantly to price changes

Pain Points:
- Poor long-term returns
- Emotional decision-making
- Addicted to short-term gains

Need:
- Discipline + risk awareness + behavior control



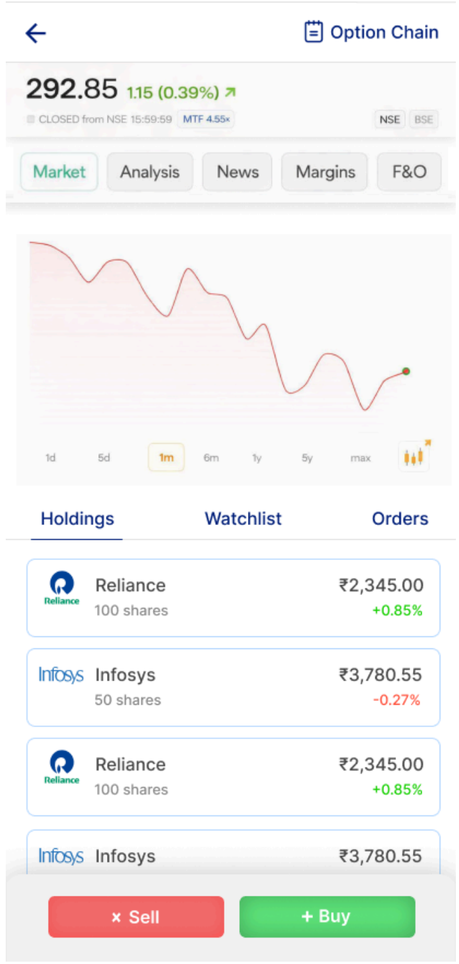
The Long-Term Investor (Bias-Drive)
Keshav Sinha
Profile:
Age: 25–45
Invests for future goals
Moderate knowledge

Behavior:
- Holds losing stocks too long
- Influenced by herd behavior
- Confused during volatility

Pain Points:
- Emotional attachment to stocks
- Inconsistent strategy
- Lack of clarity in decisions

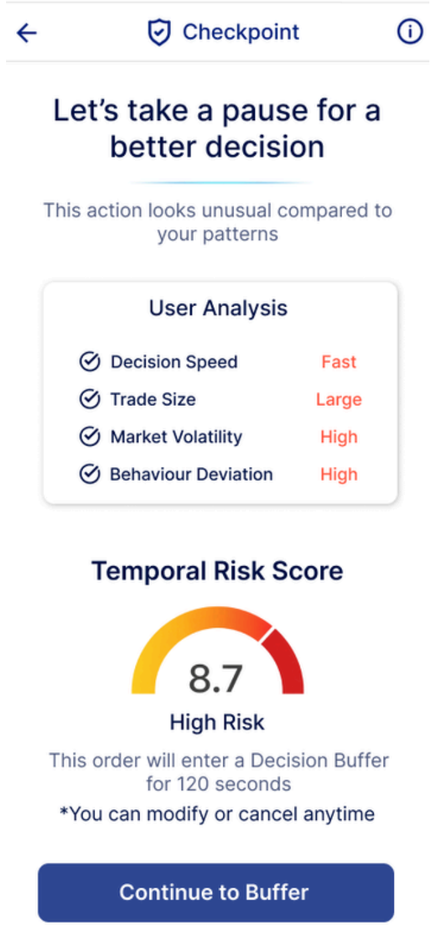
Need:
- Objective insights + bias correction + clarity

Solution Wireframes



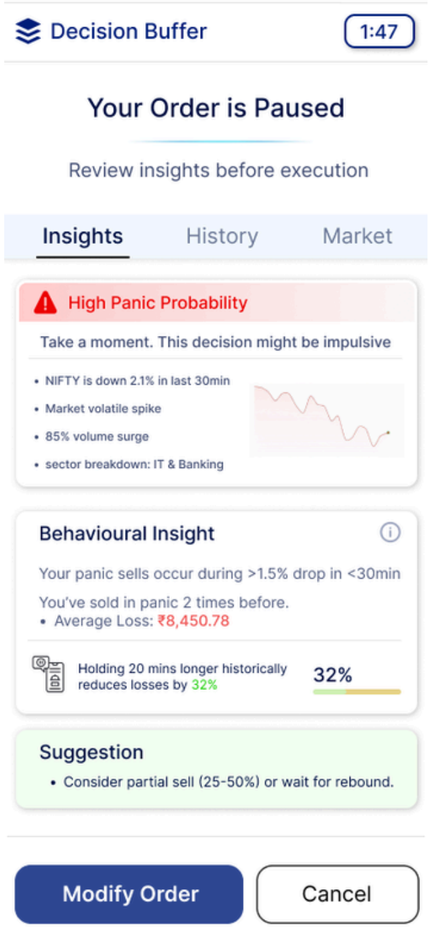
USER ACTION

User initiates a high risk action eg(sell all).



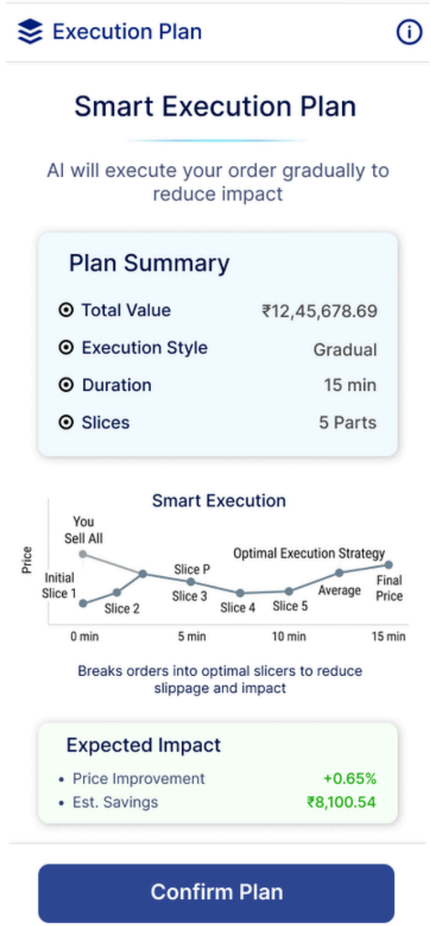
RISK ANALYSIS

Analyses intent, behaviour & market content



DECISION BUFFER

User gets time, insights and reviews back his choice.



SMART EXECUTION PLAN

Order executed in optimal slices automatically

Product Experience Flow

Step 1: User Action Initiation

When User attempts to “Sell All” during market drop or “Buy” during hype, the system captures:

- Timing
- Urgency
- Behavior pattern

Step 2: Trigger Detection

AI detects:

- Sudden action deviation
- High volatility environment
- Warns the user as:
- “High-risk emotional decision detected”

Step 3: Temporal Risk Score

System calculates a real-time decision risk score based on:

- Behavior signals (speed, frequency)
- Portfolio change
- Market conditions
- Historical patterns

Output: “Panic Probability: **HIGH**”

CORE IDEA: Instead of stopping users, our system makes them aware of the consequences of their decisions in real time

HOW WILL IT BENEFIT USER?

- Reduced impulsive decisions
- Improved long-term outcomes
- Increased user confidence

Step 4: Decision Buffer

Execution is temporarily paused
User is NOT blocked, just slowed down.
System introduces:

- Reflection window
- Smart prompts

Step 5: AI Intervention Layer

System shows:

1. Contextual Insight:
“Market dropped 8%, similar drops recovered in X months”
2. Emotional Twin:
Emotional decision → expected loss
Rational decision → expected outcome
3. Regret Simulation:
“Investors who sold in similar scenarios missed X% recovery”
4. Behavioral Feedback:
“Your current action deviates from your long-term strategy”

Step 6: Smart Execution Options

Instead of binary action:

- Sell partially
- Delay execution
- Proceed anyway

Hence, the User retains control

Technical & AI System Architecture

A real-time AI system that processes behavioral and market signals to detect emotional decision risk and trigger intelligent interventions before trade execution.

Architecture Flow

Data Layer → Feature Engineering → ML Models → Decision Engine → Intervention Layer



1. Data Input Layer

- **User Behavior Data:** clicks, trade frequency, order size, latency
- **Market Data:** price, volatility, volume, trend signals
- **User Profile:** risk appetite, goals, historical trades

2. Feature Engineering Layer

- Behavioral deviation metrics
- Volatility-adjusted indicators
- Trade timing & frequency features
- Converts raw data into model-ready feature vectors

3. AI & Decision Engine

- **Emotion Detection Model**
- **ML classification model**

Detects: fear, greed, overconfidence
Temporal Risk Scoring Engine
Computes Decision Risk Score (0–1)

Inputs:

- Behavior deviation
- Market volatility

Output:

- Probability of irrational decision
- Behavioral Pattern Model
- Learns user-specific tendencies
- Identifies recurring mistakes
- Insight Generation Engine
- Converts predictions → actionable insights

4. Intervention Layer

- Decision buffer (time-based delay)
- Real-time alerts & warnings
- Personalized suggestions
- Smart execution options
- Real-Time Feedback Loop



Business Model and Ecosystem Strategy

Our solution works as a behavioral intelligence layer that integrates with existing trading and investment platforms rather than replacing them.

Primary Revenue Model

We partner with:

- Trading platforms
- Brokerages
- Investment apps
- Wealth management platforms

Platforms pay for:

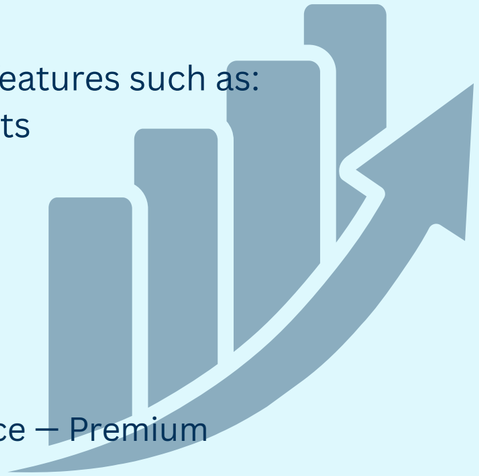
- API integration
- Per-user subscription
- Advanced behavioral analytics.

This helps platforms improve user retention, trust, and long-term engagement.

Secondary Revenue

For end users, we offer premium features such as:

- Personalized behavioral insights
- Emotional risk alerts
- Decision guidance
- Regret prediction



Freemium Model

- Basic features — Free
- Advanced behavioral intelligence — Premium

Ecosystem Strategy

Our solution connects multiple stakeholders in the financial ecosystem:

- Financial platforms (integration partners)
- Market data providers
- AI/technology infrastructure providers
- Retail investors and traders
- As more platforms adopt the system, the model improves using behavioral data, creating a network effect.

Go-to-Market Strategy



Phase 1: Partner with fintech startups

Phase 2: Integrate with established broker platforms

Phase 3: Scale as a behavioral intelligence layer across the ecosystem

ETHICS & IMPACT Building Responsible AI

CORE IDEA: The system is designed to assist, not control, ensuring users remain in charge of their financial decisions

Ethical Considerations

1. User Autonomy
- No forced blocking of trades
- Users can always override AI suggestions

2. **Transparency & Explainability**
- Every intervention includes:
- Clear reasoning
- Data-backed explanations

4. Avoiding Manipulation
- No nudging for profit-driven bias
- No influence toward specific assets

3. Data Privacy & Consent
- Explicit user consent for behavior tracking
- No misuse of personal financial data

Risks & Mitigation

Risk 1: Over-reliance on AI
- Users blindly trusting system

Mitigation:
- Encourage independent decision-making
- Provide insights, not commands

Risk 2: User Friction
- Too many interventions

Mitigation:
- Adaptive intervention (only high-risk cases)

Broader Impacts

For Investors

1. Reduced emotional losses
2. Improved financial discipline
3. Better long-term outcomes

For Financial Ecosystem

1. More stable markets
2. Reduced panicdriven volatility
3. Improved investor confidence

For AI in Finance

Shift From: Automation to Decision augmentation

Sources & References

Research Papers

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User Research

Google Form Survey on Investor Behavior:
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The Role of Emotions in Investment Decisions (With Special Reference to Palakkad, Kerala):
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Other Links

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THANKYOU!!!